# Powers of 10

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**KEY WORDS** power *(taupū)* · exponent *(taupū)* · power of 10 · expanded form

## A Positive powers of 10 ●○○

**Remember:** The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

**1.** Write each power of 10 as an ordinary number.

**a)** $10^2$ = _____ **b)** $10^3$ = _____ **c)** $10^1$ = _____ **d)** $10^5$ = _____

**e)** $10^4$ = _____ **f)** $10^0$ = _____ **g)** $10^6$ = _____ **h)** $10^8$ = _____

**2.** Write each number as a power of 10.

**a)** 1,000 = _____ **b)** 100 = _____ **c)** 100,000 = _____

**d)** 10 = _____ **e)** 1,000,000 = _____ **f)** 10,000 = _____

**g)** 1 = _____ **h)** 1,000,000,000 = _____ **i)** 10,000,000 = _____

## B Negative powers of 10 ●●○

**Watch out:** A negative power does not make the number negative. It means "one over": $10^{-2} = \frac{1}{10^2} = \frac{1}{100} = 0.01$.

**3.** Write each power of 10 as a decimal.

**a)** $10^{-1}$ = _____ **b)** $10^{-2}$ = _____ **c)** $10^{-3}$ = _____ **d)** $10^{-4}$ = _____

**4.** Write each power of 10 as a fraction.

**a)** $10^{-1}$ = _____ **b)** $10^{-3}$ = _____ **c)** $10^{-2}$ = _____ **d)** $10^{-5}$ = _____

**5.** Write each number as a power of 10.

**a)** 0.01 = _____ **b)** $\frac{1}{10}$ = _____ **c)** 0.001 = _____

**d)** $\frac{1}{10{,}000}$ = _____ **e)** 0.1 = _____ **f)** $\frac{1}{100}$ = _____

**g)** 0.00001 = _____ **h)** $\frac{1}{1{,}000{,}000}$ = _____ **i)** 0.0001 = _____



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## C Multiplying by powers of 10 ●●○

**Remember:** Multiplying by $10^2$ moves every digit 2 places to the left. Multiplying by $10^{-2}$ moves every digit 2 places to the right.

**6.** Work out.

**a)** $4.2 \times 100$ = _____ **b)** $0.37 \times 10$ = _____ **c)** $15 \times 1{,}000$ = _____ **d)** $0.56 \times 10^3$ = _____

**e)** $2.08 \times 10^2$ = _____ **f)** $0.009 \times 10^2$ = _____ **g)** $4.05 \times 10^4$ = _____ **h)** $0.6 \times 10^1$ = _____

**i)** $3.4 \times 10^{-1}$ = _____ **j)** $6 \times 10^{-2}$ = _____ **k)** $125 \times 10^{-3}$ = _____ **l)** $70 \times 10^{-1}$ = _____

**m)** $9.2 \times 10^{-2}$ = _____ **n)** $480 \times 10^{-2}$ = _____ **o)** $5 \times 10^{-3}$ = _____ **p)** $36 \times 10^0$ = _____

## D Dividing by powers of 10 ●●○

**Remember:** Dividing by $10^2$ moves every digit 2 places to the right, so the number gets smaller.

**7.** Work out.

**a)** $350 \div 10$ = _____ **b)** $78 \div 100$ = _____ **c)** $6.4 \div 10$ = _____ **d)** $29 \div 1{,}000$ = _____

**e)** $78 \div 10^2$ = _____ **f)** $0.9 \div 10^2$ = _____ **g)** $45 \div 10^3$ = _____ **h)** $8 \div 10^1$ = _____

**i)** $1{,}260 \div 10^2$ = _____ **j)** $0.3 \div 10^1$ = _____ **k)** $507 \div 10^3$ = _____ **l)** $600 \div 10^4$ = _____

## E Expanded form ●●●

**Remember:** Each digit is multiplied by the power of 10 for its place: $3.61 = 3 \times 10^0 + 6 \times 10^{-1} + 1 \times 10^{-2}$.

**8.** Write as an ordinary number.

**a)** $4 \times 10^0 + 7 \times 10^{-1}$

**b)** $2 \times 10^1 + 5 \times 10^0 + 3 \times 10^{-1}$

**c)** $6 \times 10^{-1} + 2 \times 10^{-2}$

**d)** $8 \times 10^0 + 4 \times 10^{-2}$

**e)** $3 \times 10^2 + 9 \times 10^{-1}$

**f)** $1 \times 10^2 + 3 \times 10^{-1} + 5 \times 10^{-3}$

**9.** Write each number in expanded form, using powers of 10.

**a)** 52

**b)** 3.61

**c)** 0.48

**d)** 40.06

**e)** 0.507

**f)** 7.285

## ★ Extension

**10.** Find $n$ if $10^n = 0.00001$.

**11.** How many times larger is $10^3$ than $10^{-2}$?

**12.** Work out $2.5 \div 10^{-1}$. (Think: dividing by a tenth is the same as multiplying by what?)



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NUMBER

# Powers of 10

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**KEY WORDS** power *(taupū)* · exponent *(taupū)* · power of 10 · expanded form

## A Positive powers of 10 ●○○

**Remember:** The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

**1.** Write each power of 10 as an ordinary number.

**a)** $10^3$ = _____ **b)** $10^2$ = _____ **c)** $10^4$ = _____ **d)** $10^1$ = _____

**e)** $10^6$ = _____ **f)** $10^0$ = _____ **g)** $10^5$ = _____ **h)** $10^7$ = _____

**2.** Write each number as a power of 10.

**a)** 100 = _____ **b)** 10,000 = _____ **c)** 1,000 = _____

**d)** 10 = _____ **e)** 100,000 = _____ **f)** 10,000,000 = _____

**g)** 1 = _____ **h)** 100,000,000 = _____ **i)** 1,000,000 = _____

## B Negative powers of 10 ●●○

**Watch out:** A negative power does not make the number negative. It means "one over":
$10^{-2} = \frac{1}{10^2} = \frac{1}{100} = 0.01$.

**3.** Write each power of 10 as a decimal.

**a)** $10^{-2}$ = _____ **b)** $10^{-1}$ = _____ **c)** $10^{-4}$ = _____ **d)** $10^{-3}$ = _____

**4.** Write each power of 10 as a fraction.

**a)** $10^{-2}$ = _____ **b)** $10^{-1}$ = _____ **c)** $10^{-4}$ = _____ **d)** $10^{-3}$ = _____

**5.** Write each number as a power of 10.

**a)** 0.1 = _____ **b)** $\frac{1}{100}$ = _____ **c)** 0.0001 = _____

**d)** $\frac{1}{1{,}000}$ = _____ **e)** 0.01 = _____ **f)** $\frac{1}{10}$ = _____

**g)** 0.000001 = _____ **h)** $\frac{1}{100{,}000}$ = _____ **i)** 0.001 = _____


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## C Multiplying by powers of 10 ●●○

**Remember:** Multiplying by $10^2$ moves every digit 2 places to the left. Multiplying by $10^{-2}$ moves every digit 2 places to the right.

**6.** Work out.

**a)** $5.3 \times 100$ = _____ **b)** $0.48 \times 10$ = _____ **c)** $27 \times 1{,}000$ = _____ **d)** $0.72 \times 10^3$ = _____

**e)** $6.04 \times 10^2$ = _____ **f)** $0.007 \times 10^2$ = _____ **g)** $3.06 \times 10^4$ = _____ **h)** $0.9 \times 10^1$ = _____

**i)** $5.8 \times 10^{-1}$ = _____ **j)** $4 \times 10^{-2}$ = _____ **k)** $375 \times 10^{-3}$ = _____ **l)** $90 \times 10^{-1}$ = _____

**m)** $7.4 \times 10^{-2}$ = _____ **n)** $650 \times 10^{-2}$ = _____ **o)** $8 \times 10^{-3}$ = _____ **p)** $52 \times 10^0$ = _____

## D Dividing by powers of 10 ●●○

**Remember:** Dividing by $10^2$ moves every digit 2 places to the right, so the number gets smaller.

**7.** Work out.

**a)** $470 \div 10$ = _____ **b)** $63 \div 100$ = _____ **c)** $8.2 \div 10$ = _____ **d)** $37 \div 1{,}000$ = _____

**e)** $94 \div 10^2$ = _____ **f)** $0.5 \div 10^2$ = _____ **g)** $28 \div 10^3$ = _____ **h)** $6 \div 10^1$ = _____

**i)** $2{,}340 \div 10^2$ = _____ **j)** $0.7 \div 10^1$ = _____ **k)** $409 \div 10^3$ = _____ **l)** $900 \div 10^4$ = _____

## E Expanded form ●●●

**Remember:** Each digit is multiplied by the power of 10 for its place: $3.61 = 3 \times 10^0 + 6 \times 10^{-1} + 1 \times 10^{-2}$.

**8.** Write as an ordinary number.

**a)** $6 \times 10^0 + 3 \times 10^{-1}$

**b)** $4 \times 10^1 + 7 \times 10^0 + 2 \times 10^{-1}$

**c)** $8 \times 10^{-1} + 5 \times 10^{-2}$

**d)** $5 \times 10^0 + 7 \times 10^{-2}$

**e)** $5 \times 10^2 + 4 \times 10^{-1}$

**f)** $2 \times 10^2 + 1 \times 10^{-1} + 6 \times 10^{-3}$

**9.** Write each number in expanded form, using powers of 10.

**a)** 84

**b)** 2.95

**c)** 0.73

**d)** 60.02

**e)** 0.309

**f)** 4.618

## ★ Extension

**10.** Find $n$ if $10^n = 0.000001$.

**11.** How many times larger is $10^2$ than $10^{-3}$?

**12.** Work out $3.6 \div 10^{-2}$. (Think: dividing by a tenth is the same as multiplying by what?)



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